Math 526, S12 Name: Answer

Quiz 3 Section: SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Given a system of euqations

$$\begin{aligned} x + 2y + 3z &= 1 \\ 2x + 4y - z &= 4 \\ 3y + z &= 3 \end{aligned}$$

(a) Write down the augmented matrix $[\mathbf{A}, \mathbf{b}]$.

$$[A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{array}\right]$$

(b) Find the *Elimination matrix* $\mathbf{E}_{21}$ which can be used to eliminate the the first component below the first pivot. Calculate $\mathbf{E}_{21}[\mathbf{A}, \mathbf{b}]$.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E_{21}[A, b] = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{array}\right]$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & -7 & 2 \\ 0 & 3 & 1 & 3 \end{array}\right]$$

(c) Find the *Permutation matrix* $\mathbf{P}_{23}$ such that $\mathbf{P}_{23}\mathbf{E}_{21}[\mathbf{A}, \mathbf{b}]$ becomes triangular.

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\left(P_{23} E_{21} [A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 3 & 1 & 3 \\ 0 & 0 & -7 & 2 \end{array}\right] \right)$$

Problem 2. (4 points) Let

$$\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

(a) Compute $\mathbf{A}^2$ and $\mathbf{A}^3$.

$$A^2 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 3 & 1 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 0 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 \\ 7 & 1 \end{bmatrix}$$

(b) Predict $\mathbf{A}^n$ based on results from (a).

$$A^n = \begin{bmatrix} 2^n & 0 \\ 2^n - 1 & 1 \end{bmatrix}$$